

Kunj Rathod

AI Systems Engineer — Coding Agents, Agent Harnesses, Evals, Developer Tooling & Reliable LLM Systems

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Education

University of Utah — B.S. Computer Science, GPA 3.7/4.0, Dean's List

Aug 2023 – Dec 2026

Coursework: Distributed Systems, Operating Systems, Machine Learning, Computer Vision, NLP, Algorithms, Databases

Experience

Software Engineering Intern — Microsoft Fabric (Azure Data)

May 2026 – Aug 2026

Microsoft, Redmond, WA

- Building SKU Change History end-to-end for the Fabric Manage Capacities admin portal: query builder, service layer, REST API, data contracts, and telemetry-backed audit timelines across enterprise tenants.
- Migrating capacity overage workflows into native Fabric platform UX; partnering across UX/backend teams to turn ambiguous product/debugging needs into durable platform behavior and observable admin workflows.

Software Development Intern, AI Services (SUDO Program)

Jan 2025 – Present

University of Utah Health, Salt Lake City, UT

- Architected HIPAA-compliant LLM serving infrastructure for 90+ hospital executives: AWS Bedrock Agents, Knowledge Bases, Guardrails, streaming inference, API caching, and session persistence for 1,000+ conversations.
- Reduced latency 40% with p95 time-to-first-token <200ms; owned API design, schema, observability-minded debugging, and 6 production features across React/TypeScript, Flask, Lambda, DynamoDB, S3, and RDS.

AI Engineering Intern

Nov 2024 – Apr 2025

CourtEasy.ai / Nugen, Remote

- Scaled hybrid RAG over 10M+ legal documents using dense retrieval, BM25, and reranking; improved recall 28%, reduced hallucinations 35%, and served 5,000+ daily legal-research queries.
- Benchmarked 8 LLM families on accuracy, latency, and cost; built model-routing analysis projected to cut inference spend by \$50k+/yr while preserving answer quality.

Agent Systems, Evals & Research

MEMROT v2 — Mechanistic Evaluation of Long-Context Memory Failure

May 2026 – Present

Independent research artifact; Gemma-2-2B, Gemma Scope SAEs, LongMemEval-S, CHPC/SLURM

- Built an ablation-heavy evaluation harness for controlled haystack retrieval: gold-position randomization, in-hook attention reduction, empirical SAE layer scans, mean-ablation/intervention tests, and an 840-pass sweep over 140 exactly-one-gold questions.
- Separated model behavior from measurement artifacts with explicit limitations around feature selection, attention normalization, and heuristic grading; prioritized reproducible negative/causal evidence over inflated claims.

Long-term Active Embodied QA (LA-EQA)

Spring 2026

University of Utah; embodied agents, VLM video QA, episodic memory, Qwen3-VL

- Built Video Mind Palace, a sampled-frame VLM baseline replacing Robotic Mind Palace's offline GPT-4o scene-graph pipeline; achieved 31–57% online inference speedup at 13–17% relative accuracy tradeoff.
- Audited failure modes in the 100-question benchmark and proposed SceneSmith + MuJoCo environments for longer, interactive, multi-agent embodied QA evaluation.

Selected Projects

Continuum / FNDR — Local-First Multimodal Agent Memory Engine

2024 – Present

Rust, Tauri 2, React, TypeScript, LanceDB, Apple Vision, OCR, embeddings, MCP

- Built a privacy-first desktop memory layer that captures foreground context on-device into vector + lexical retrieval fields, reconstructs temporal context from screen/audio/web activity, and exposes agent-ready recall through an MCP server.

HirePilot — 9-Agent Autonomous Recruiting Pipeline

Mar 2026

Podium AI Hackathon Winner; Python/FastAPI, Anthropic, Perplexity, GitHub REST, Twilio, Slack, Google Calendar

- Shipped a real-integration agent workflow in 24 hours: GitHub sourcing, enrichment, Anthropic ranking, live Twilio voice interviews, transcript analysis, Slack approval controls, and calendar scheduling.

Technical Skills

Languages: Python, Rust, TypeScript, C++, Java, Swift, SQL Systems: Docker, Kubernetes, Linux/macOS tooling, REST APIs, CI/CD, SLURM, gRPC

AI Systems: coding agents, agent orchestration, tool-using LLMs, evals/ablations, safe tool workflows, RAG/GraphRAG, vector search, BM25, reranking, MCP, PyTorch, TensorFlow, VLMs

Cloud/Data: AWS Bedrock, Lambda, API Gateway, S3, DynamoDB, RDS, Azure, FastAPI, Flask, NebulaGraph, LanceDB, PostgreSQL